

Missing Data Mechanism

Defn:

Cause

Types:

MAR

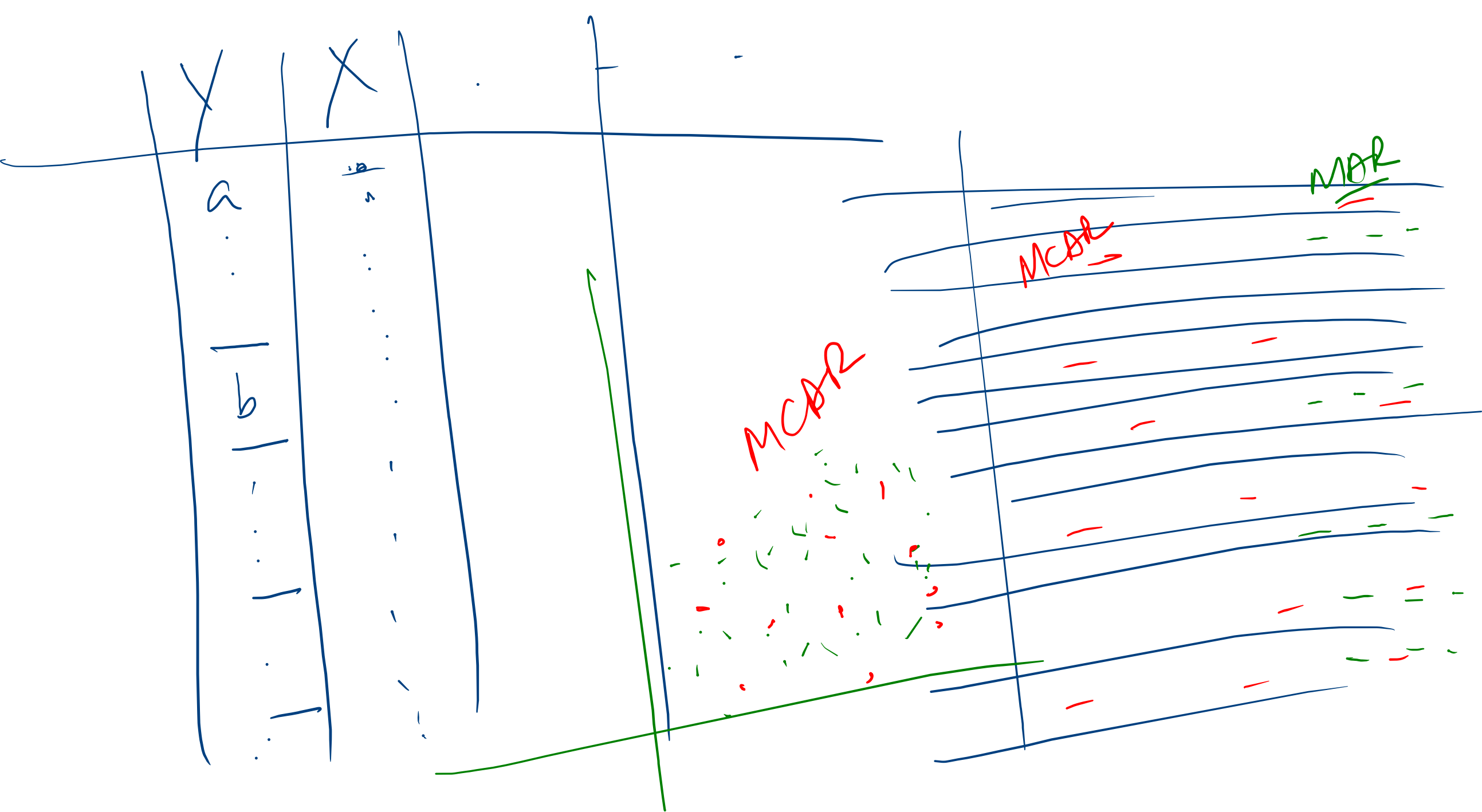
MAR

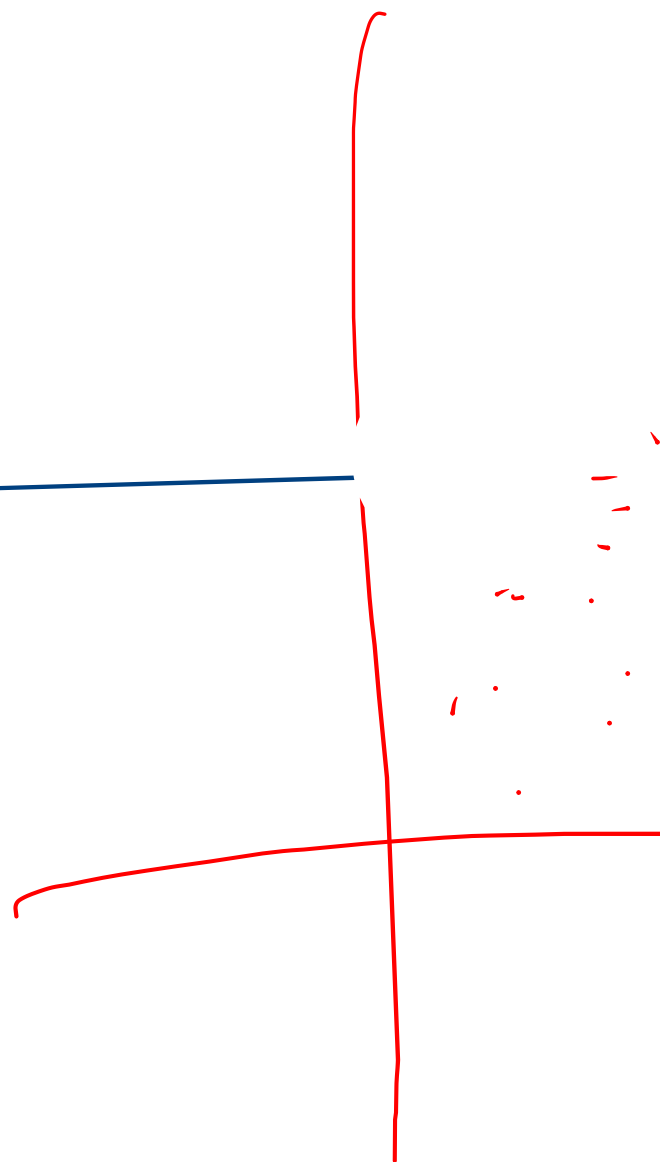
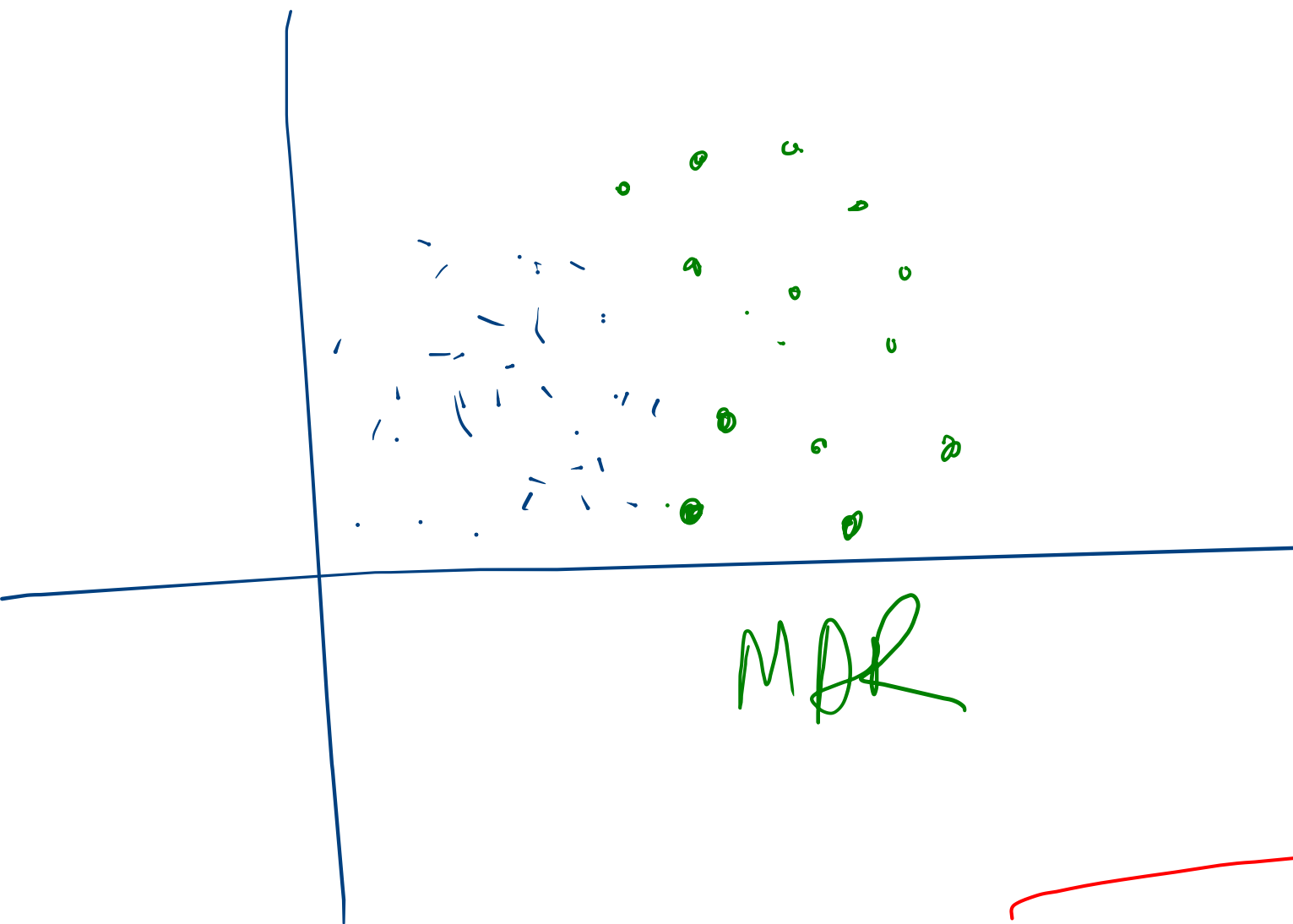
MNAR

Missing values
doesn't depend on the
observed values or missing values

Missing values
depend on
obs. values.

Missing values
depend
on
the
missing
values
itself





Detection

MCAR

Numerical



describe
isnull
isna

Visual



histogram
Scatter plot

obs. values

replaced

missingno

Missingno Package

install

import

missingno

as msno

missingno

dendrogram

heatmap

matrix

bar

Y	R	Y _{mis}	Y _{obs}
0	1		
0	1		
.	1		
0	0		
0	0		
0	0		
0	0		

$$R = \begin{cases} 0 & \text{if value missing} \\ 1 & \text{if value observed} \end{cases}$$

$$Y = \begin{pmatrix} Y_{obs} & Y_{mis} \end{pmatrix}$$

: parameters

~~MIM in MCAR~~

$$P(R | Y_{\text{obs}}, Y_{\text{mis}}, \psi)$$

$$\underline{\underline{R=0}}$$

 Missing

$$P(R=0 | Y_{\text{obs}}, Y_{\text{mis}}, \psi) = P(R=0)$$

MAR

$$P(R=0 \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi) = P(R=0 \mid Y_{\text{obs}}, \underline{\psi})$$

MNAR

$$P(R=0 \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi) =$$